

Drilling Fluid Program: 12-1/4 in Hole Section (4,120–4,450 m MD) (illustrative)

Document ID: MP-MN-SM-DW-01-SEC4 | Type: MUD_PROGRAM | Well: MN-SM-DW-01 | Date: 2026-08-01

1. Section Details & Mud Specifications

- **Hole Size**: 12-1/4 in
- **Interval**: 3,850 m to 4,450 m MD (Section TD)
- **Base Mud**: Low-toxicity Synthetic Oil-Based Mud (SOBM, 80/20 Oil/Water ratio).
- **Initial Mud Density**: 11.20 ppg (1.34 SG).
- **Contingency Weighted Density**: 11.65 ppg (1.40 SG).
- **Target Plastic Viscosity**: 26–30 cP.
- **Target Yield Point**: 16–20 lb/100 ft².

2. Barite Weighting Protocol

- At 4,172 m MD, raise active system density to 11.65 ppg using 87,700 lb API barite.
- Active system volume: 3,100 bbl; Riser volume: 1,950 bbl.
- Verify zero sag and mud weight parity before entering sand at 4,195 m MD.